Name- Shreyash Khapekar

PRN - B25ET1019

Division- SY1 ENTC

Title: Represent graphs and generate minimum spanning trees using Prim's algorithm.

```
#include <stdio.h>

#define INF 999

int main(){
   int n, i, j, u, v;
   int cost[10][10], visited[10] = {0};
   int edges = 0, minCost = 0;

   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter cost matrix (0 for no edge):\n");
   for(i = 0; i < n; i++){
      for(j = 0; j < n; j++){
         scanf("%d", &cost[i][j]);
         if(cost[i][j] == 0)
            cost[i][j] = INF;
      }
   }

   visited[0] = 1;

   printf("\nEdges in Minimum Spanning Tree:\n");

   while(edges < n - 1){
      int min = INF;
      u = -1;
      v = -1;

      for(i = 0; i < n; i++){
         if(visited[i] == 1){
            for(j = 0; j < n; j++){
               if(visited[j] == 0 && cost[i][j] < min){
```

```
                min = cost[i][j];
                u = i;
                v = j;
            }
        }
    }
}

if(u != -1 && v != -1){
    printf("Edge %d - %d : Cost %d\n", u, v, min);
    minCost = minCost + min;
    visited[v] = 1;
    edges++;
}
}

printf("Total Cost of Minimum Spanning Tree = %d\n", minCost);

return 0;
}
```